

8. Annex

8.1. Table of the datasets included in HARIS

This table lists the geo-located datasets, currently, inserted in the repository/database of HARIS. Nevertheless, other information could be included or modified in the next months of the project (e.g. Socio-Economic indicators not yet included).

Domain	Element	ARCH cities			
		Bratislava	Camerino	Hamburg	Valencia
Natural Environment	Vegetation	X	X	-	X
	Geology	-	X	X	X
	Water and waterways	X	X	X	X
Built Environment	Buildings	X	X	X	X
	Road infrastructures (Roads, Railways, etc.)	X	X	X	X
	3D City Model	X*	X*	X**	-
Cultural Heritage Assets	Construction (and Objects)	X	X (X)	X	
	Crops				X
Social structure	Social indicators (population, education, etc.)	X ⁺	X ⁺		X ⁺⁺
Economic status	Economic indicators (activities, employed people)		X ⁺		X ⁺⁺
Community (Facilities)	Hospital and health system, schools, university, etc.	X	X	X	X
	Other public facilities (museums, theatres, etc.)	X	X	X	X
	Emergency services (firefighters, police, etc.)	X	X	X	X
Functional areas	Land Use\Land Cover	X [°]	X ^{°°}	X ^{°°}	X ^{°°}

* 3D model of HA – Level of Details (LoD) 1 by CityGML; **3D model of the whole city – LoD2 by CityGML

⁺ dataset related to the last census (2011); ⁺⁺ last available dataset (2020)

[°] CORINE Land Cover inventory (<https://land.copernicus.eu/pan-european/corine-land-cover>)

^{°°} Urban Atlas inventory (<https://land.copernicus.eu/local>)

8.2. Tables of the Construction\Object DB

This annex represents the guide to all the tables, with the relative fields, that make up the database of HARIS, according to the structures showed in sections 3.1.1 and 3.1.2 both for constructions (section 8.2.1) and objects (section 8.2.2). In particular, the fields that can be modified through the Construction\Object Sheet (as indicated in section 4.3) are described, as well as those that define the values that some of the fields can assume, and the others that allow the connection between the information.

8.2.1. Tables of the CONSTRUCTION database

8.2.1.1. CONSTRUCTION

Attribute	Value Type	Definition
OBJECTID	ObjectID	ID of the element in the table
BeginLifespanVersion	DateTime	Date and time at which this version of the spatial object was inserted or changed in the spatial data set.
EndLifespanVersion	DateTime	Date and time at which this version of the spatial object was superseded or retired in the spatial data set.
ConditionOfConstructionRef	ConditionOfConstructionValue	see label in ConditionOfConstructionValue table
DateOfConstruction	DateOfEvent	Date of construction.
DateOfDemolition	DateOfEvent	Date of demolition.
DateOfRenovation	DateOfEvent	Date of last major renovation.
Name	GeographicalName	Name of the asset
LegalConditions	Ineger	Public, Private, Public private partnership
Place	Text	Place
Address	Text	Address
CityRef	Integer	Reference to the Cities
Sheet	Integer	Cadastral Sheet
Parcel	String	Cadastral Parcel
SubA	String	Cadastral subaltern\subordinate

8.2.1.2. CONSTRUCTION_TO_ELEVATION

Attribute	Value Type	Definition
OBJECTID	ObjectID	ID of the element in the table
ConstructionId	integer	Reference to the CONSTRUCTION table
ElevationId	Integer	Reference to the ELEVATION table

8.2.1.3. CONSTRUCTION_TO_EXTERNAL_REFERENCE

Attribute	Value Type	Definition
OBJECTID	ObjectID	ID of the element in the table
ConstructionRef	Integer	Reference to the CONSTRUCTION table
ExternalReferenceRef	Integer	Reference to the EXTERNAL_REFERENCE table

8.2.1.4. BUILDING

Attribute	Value Type	Definition
OBJECTID	ObjectID	ID of the element in the table
CurrentUse	Current use	Activity hosted within the building. This attribute addresses mainly the buildings hosting human activities
NumberOfBuilding Units	Integer	Number of building units in the building. A BuildingUnit is a subdivision of Building with its own lockable access from the outside or from a common area (i.e. not from another BuildingUnit), which is atomic, functionally independent, and may be separately sold, rented out, inherited, etc.
NumberOfDwellings	Integer	Number of dwellings
NumberOfFloorsAboveGround	Integer	Number of floors above ground
ConstructionRef	Integer	Reference to the CONSTRUCTION table

8.2.1.5. BUILDING_GEOMETRY_2D

Attribute	Value Type	Definition
OBJECTID	ObjectID	ID of the element in the table
BuildingRef	Integer	Reference to the BUILDING table
HorGeomEstimated Accuracy	Real	The estimated absolute positional accuracy of the (X,Y) coordinates of the building geometry. Absolute positional accuracy is defined as the mean value of the positional uncertainties for a set of positions where the positional uncertainties are defined as the distance between a measured position and what is considered as the corresponding true position.
HorGeomReference	HorizontalGeometryReferenceValue	Element of the building that was captured by (X,Y) coordinates.
ReferenceGeometry	Boolean	The geometry to be taken into account by view services, for portrayal.
VerGeomEstimated Accuracy	Real	The estimated absolute positional accuracy of the Z coordinates of the building geometry. Absolute positional accuracy is defined as the mean value of the positional uncertainties for a set of positions where the positional uncertainties are defined as the distance between a measured position and what is considered as the corresponding true position.
VerGeomReference	ElevationReferenceValue	Element of the building that was captured by vertical coordinates.

8.2.1.6. BUILDING_NATURE

Attribute	Value Type	Definition
OBJECTID	ObjectID	ID of the element in the table
BuildingNatureLabel	BuildingNatureType	Values indicating the nature of a building.
BuildingNatureDescription	BuildingNatureDescription	Description of the nature of a building.

8.2.1.7. BUILDING_TO_BUILDING_NATURE

Attribute	Value Type	Definition
OBJECTID	ObjectID	ID of the element in the table
BuildingRef	integer	Reference to the BUILDING table
BuildingNatureRef	integer	Reference to the BUILDING_NATURE table

8.2.1.8. CURRENT_USE

Attribute	Value Type	Definition
OBJECTID	ObjectID	ID of the element in the table
CurrentUseValueRef	CurrentUseValue	Reference to the CURRENT_USE_VALUE table
Occupants	Integer	Number of potential occupants
Percentage	Integer	The proportion of the real world object, given as a percentage, devoted to this current use.

8.2.1.9. CURRENT_USE_VALUE

Attribute	Value Type	Definition
OBJECTID	ObjectID	ID of the element in the table
CurrentUseValueLabel	CurrentUseValue_Label	Refer to CurrentUseValue table
CurrentUseValueDescription	CurrentUseValue_Description	Refer to CurrentUseValue table

8.2.1.10. STRUCTURE

Attribute	Alias	Value Type	Definition
OBJECTID		ObjectID	ID of the element in the table
ConstructionRef		integer	Reference to the CONSTRUCTION table
Name	Name	text	Identification
Additional notes	Notes	text	notes on the considered building
B1	Nr Storeys	integer	Number of storeys (including basements)
B2	Nr Basements	integer	Number of basements
B3	Average storey height [m]	integer	Average storey height
B4	Average storey surface [m]	integer	Average storey surface
B5	Construction/renovation period	Integer (1..8)	Period of construction or period relative to last important renovation [1=<1919, 2=1919-1945, 3=1946-1961, 4=1962-1971, 5=1972-1981, 6=1982-1991, 7=1992-2001, 8=>2002]
B6	Nr storeys above ground	integer	Number of storeys above ground

Attribute	Alias		Value Type	Definition
B7	Position Aggregate	in	integer	Building position (in the aggregate) 0=isolated 1=internal building, 2=extreme building, 3=corner building 4=other
B8	Regularity plan	in	Integer	1 = Not Regular 2 = Regular
B9	Structure		Integer	0 = Masonry 1 = Reinforced Concrete Frame 2 = Reinforced Concrete Wall 3 = Steel Frame 4 = Wood Beams 5 = Iron Beams 6 = Cast Iron Coated
C0_1	Structure Typology T1 [%]		Real	Percentage of the prevailing typology of vertical/horizontal structures. The following fields contain information about the building typology according to AEDEs form (post-earthquake survey). Reference document https://www.eeri.org/wp-content/uploads/Italy/EUR%2022868%20(2007)%20Field%20Manual%20for%20post-earthquake%20damage%20assessment.pdf
C0_2	T1-Vertical Structures		integer	Vertical Structures (prevailing typology) 1=unknown 2=Irregular layout or bad quality/without tie rods or tie beams 3=Irregular layout or bad quality/with tie rods or tie beams 4=Regular layout and good quality/ without tie rods or tie beams 5=Regular layout and good quality/with tie rods or tie beams
C0_3	T1-Horizontal Structures		Integer	C0-3 T1-Horizontal Structures integer (1,...,6) Horizontal structures (prevailing typology) 1=Unknown 2=Vaults without tie rods 3=Vaults with tie rods 4=Beams with flexible slab 5=Beams with semirigid slab 6=Beams with rigid slab) X
C0_4	Structure Typology T2 [%]		Real	Percentage of the secondary typology (derived as 100-C0-1) of vertical/horizontal structures

Attribute	Alias	Value Type	Definition
C0_5	T2-Vertical Structures	Integer	Vertical Structures (secondary typology) 1=unknown 2=Irregular layout or bad quality/without tie rods or tie beams 3=Irregular layout or bad quality/with tie rods or tie beams 4=Regular layout and good quality/ without tie rods or tie beams 5=Regular layout and good quality/with tie rods or tie beams 6 Structures (secondary typology)
C0_6	T2-Horizontal Structures	Integer	Horizontal structures (secondary typology) 1=Unknown 2=Vaults without tie rods 3=Vaults with tie rods 4=Beams with flexible slab 5=Beams with semirigid slab 6=Beams with rigid slab) structures (secondary typology)
C0_7	Isolated columns	Integer	Presence of isolated columns 0=no, 1=yes
C0_8	Mixed structures	Integer	Presence of a mixed structure typology 1=frame over masonry 2=masonry over frame 3=masonry and frame in parallel
C0_9	Strengthening	Integer	Presence of strengthened masonry 1=injections or unreinforced plasters 2=reinforced masonry of reinforced plasters 3=other or unknown strengthening
C0_10	Roof	Integer	Roof 1=thrusting heavy 2=non thrusting heavy 3=thrusting light 4=non thrusting light
C1_1	Material M1 [%]	Real (0-100)	Percentage related to the prevailing typology The following fields contain information about the masonry survey campaign. There are 2 prevalent types that will be weighted according to the percentage of presence. (reference document to be included)
C1_2	M1-Masonry Typo	Integer (1..7)	Prevailing typology Code for field 2.2 TYPOLOGY in MQI Form 1=stone 2=brick 3=semi-hollow bricks (<45%) 4=hollow bricks (>45%) 5=tuffs elements 6=mixed masonry 7=other

Attribute	Alias	Value Type	Definition
C1_3	M1-Brick courses	Integer (1,...,6)	Prevailing typology Code for field 2.3 BRICK COURSES in MQI Form 1=present at a fixed distance 2=present at variable distance 3=present (other material) 4=present (brick) 5=present (RC) 6=absent
C1_4	M1-Listed Masonry	Integer (1,...,2)	Prevailing typology Code for field 2.4 LISTED MASONRY in MQI Form 1=absent 2=present
C1_5	M1-pinning stones	Integer (1,...,3)	Prevailing typology Code for field 2.5 PINNING STONES in MQI Form 1=absent 2=present (brick) 3=present (stone)
C1_6	M1-Horizontal Joint	Integer (1,...,3)	Prevailing typology Code for field 2.6 HORIZONTALITY OF BED JOINT in MQI Form 1=not fulfilled 2=partially fulfilled 3=fulfilled
C1_7	M1-Vertical Joint	Integer (1,...,3)	Prevailing typology Code for field 2.7 STAGGER PROPERTIES OF VERTICAL JOINTS in MQI Form 1=Aligned vertical joint (NF) 2=Partially staggered vertical joints (PF) 3=Properly staggered vertical joints (F)
C1_8	M1-Elements	Integer (1,...,8)	Prevailing typology Code for field 3.1.1 STONE/BRICK ELEMENT in MQI Form 1=sandstone 2=limestone 3=hardstone 4=tuff 5=bricks 6=unfired bricks 7=RC 8=other
C1_9	M1-Origin	Integer (1,...,5)	Prevailing typology Code for field 3.1.2 ORIGIN OF THE STONE/BRICK in MQI Form 1=local excavation 2=river bed 3=quarry 4=artificial product 5=other

Attribute	Alias	Value Type	Definition
C1_10	M1-Processing	Integer (1,...,4)	Prevailing typology Code for field 3.1.3 STONE/BRICK PROCESSING in MQI Form 1=no processing 2=rounded or pebble stonework 3=natural square block 4=artificial square block
C1_11	M1-Conservation	Integer (1,...,4)	Prevailing typology Code for field 3.1.4 CONSERVATION STATE OF STONE/BRICK in MQI Form 1=good 2=decent 3=poor 4=very bad
C1_12	M1-Shape	Integer (1,...,7)	Prevailing typology Code for field 3.1.5a STONE/BRICK SHAPE in MQI Form 1=pebbles 2=erratic blocks 3=slabs 4=rubble 5=artificial square blocks 6=blocks 7=square blocks
C1_13	M1-Regularity Class	Integer (1,...,3)	Prevailing typology Code for field 3.1.5b STONE/BRICK REGULARITY in MQI Form 1=Not fulfilled 2=Partially fulfilled 3=Fulfilled
C1_14	M1-Dimension	Integer (1,...,3)	Prevailing typology Code for field 3.1.6a STONE/BRICK DIMENSION in MQI Form 1= stones dimension < 20cm 2=stones dimension 20-40 cm 3=stones dimension > 40 cm
C1_15	M1-Dimension Class	Integer (1,...,3)	Prevailing typology Code for field 3.1.6b STONE/BRICK DIMENSION REGULARITY in MQI Form 1=Not fulfilled 2=Partially fulfilled 3=Fulfilled
C1_16	M1-Mortar function	Integer (1,...,3)	Prevailing typology Code for field 3.2.1 MORTAR FUNCTION in MQI Form 1=bedding 2=facework 3=filling

Attribute	Alias	Value Type	Definition
C1_17	M1-Mortar Consistency	Integer (1,...,5)	Prevailing Code for field 3.2.2 MORTAR CONSISTENCY in MQI Form 1=dusty 2=weak 3=medium 4=good 5=other quality quality typology mortar mortar mortar mortar
C1_18	M1.Mortar Colour	Integer (1,...,12)	Prevailing Code for field 3.2.3 MORTAR COLOR in MQI Form 1=whitish 2=ivory 3=light 4=grey 5=dark 6=beige 7=grey 8=ochre 9=hazelnut 10=light 11=brown 12=pinkish – white grey grey ochre brown
C1_19	M1-Mortar Aggregate Colour	Integer (1,...,5)	Prevailing Code for field 3.2.4 (MORTAR) COLOR OF THE AGGREGATE in MQI Form 1=whitish 2=ivory 3=light 4=grey 5=dark 6=beige 7=grey 8=ochre 9=hazelnut 10=light 11=brown 12=pinkish – white grey grey ochre brown
C1_20	M1-Mortar Aggregate Type	Integer (1,...,3)	Prevailing Code for field 3.2.5 (MORTAR) TYPE OF AGGREGATE in MQI Form 1=sand 2=gravel 3=fine gravel typology
C1_21	M1-Mortar Aggregate Shape	Integer (1,...,2)	Prevailing Code for field 3.2.6 (MORTAR) SHAPE OF THE AGGREGATE in MQI Form 1=rounded 2=edgy typology

Attribute	Alias	Value Type	Definition
C1_22	M1-Joint Regularity	Integer (1,...,3)	Prevailing Code for field 3.2.7° (MORTAR) BED JOINT typology REGULARITY in MQI Form 1=Not Fulfilled 2=Partially Fulfilled 3=Fulfilled
C1_23	M1-Joint Dimension	Integer (1,...,2)	Prevailing Code for field 3.2.7b (MORTAR) BED JOINTS typology DIMENSION in MQI Form 1=thin 2=thick
C1_24	M1-Mortar Quality Class	Integer (1,...,3)	Prevailing Code for field 3.2.7 QUALITY OF THE MORTAR in typology MQI Form 1=Not Fulfilled 2=Partially Fulfilled 3=Fulfilled
C1_25	M1-NTC Class	Integer (1,...,8)	Prevailing NTC 2019 - Circolare classification, Tab. C8-5-1 typology 1=Irregular stone masonry (pebbles, erratic, irregular stones) 2=Uncut stone masonry with facing walls of limited thickness and infill core 3=Cut stone with good bonding 4=Irregular soft stone masonry (tuff, limestone, etc.) 5=Regular soft stone masonry (tuff, limestone, etc.) 6=Dressed rectangular (ashlar) stone masonry 7=Solid brick masonry with lime mortar 8=Semi-solid brick masonry with cement mortar

8.2.1.11. CONSTRUCTION MATERIAL

Attribute	Alias	Value Type	Definition
OBJECTID		ObjectID	ID of the element in the table
ConstructionRef		integer	Reference to the CONSTRUCTION table
D0_1	Prevalent Material	text	Type of prevalent material of the construction
D0_2	Plaster	Real (0..1)	Percentage of Plaster
D0_3	Sandstone	Real (0..1)	Percentage of Sandstone
D0_4	Limestone	Real (0..1)	Percentage of Limestone
D0_5	Plaster	Real (0..1)	Percentage of plaster
D0_6	Brick	Real (0..1)	Percentage of Brick
D0_7	Other Materials	Real (0..1)	Percentage of Other Materials

Attribute	Alias	Value Type	Definition
D1_1	M1-Carbonate Content	Integer (0,1)	Prevailing typology (according to typology considered in C1-1 – C1-25) Quality index relative to the chemical-physical parameter n.1 CARBONATE CONTENT BY CALCIMETRY 0= 0-10% 1=>10%
D1_2	M1-Salt Content	Integer (1..5)	Prevailing typology Quality index relative to the chemical-physical parameter n.2 SOLUTE SALT CONTENT BY CONDUCTIMETRY, measured according to UNI11087 and classified as follow 1=(>20%), 2=(10-20%), 3=(10-6%), 4=(3-6%), 5=(<3%)
D1_3	M1-Porosity	Integer (1..3)	Prevailing typology Quality index relative to the chemical-physical parameter n.3 MORTAR POROSITY BY PHISISORPTION POROSIMETRY 1=(>40%) 2= (20-40%), 3=(<20%)
D1_4	M1-Granulometry	Integer (1..4)	Prevailing typology Quality index relative to the chemical-physical parameter n.4 GRANULOMETRIC ANALYSIS BY THIN SECTIONS, classified according to CNR NorMsL12/83 guidelines (in Italian) 1= homogeneous (1 class) 2= almost homogeneous (2 classes) 3= almost heterogeneous (3 classes) 4= heterogeneous (>3 class)
D1_6	M1-Dimension MQI	Vector (dim.3) of reals (0..10)	Prevailing typology MQI score for STONE/BRICK DIMENSION (values between 0 and 10, 3 fields for vertical, orthogonal, coplanar behaviour)
D1_7	M1-Strenght MQI	Vector (dim.3) of reals (0..10)	Prevailing typology MQI score for STONE/BRICK STRENGHT (values between 0 and 10, 3 fields for vertical, orthogonal, coplanar behaviour)
D1_8	M1-Mortat MQI	Vector (dim.3) of reals (0..10)	Prevailing typology MQI score for QUALITY OF MORTAR (values between 0 and 10, 3 fields for vertical, orthogonal, coplanar behaviour)
D1_9	M1-Masonry MQI	Vector (dim.3) of reals (0..10)	Prevailing typology MQI score for MASONRY(values between 0 and 10, 3 fields for vertical, orthogonal, coplanar behaviour)

Attribute	Alias	Value Type	Definition
D1_10	M1-Masonry Category	Vector (dim.2) integers (1..3) of	Prevailing typology MASONRY overall CATEGORY 1=category A, 2=B, 3=C, 3 fields for vertical, orthogonal, coplanar behaviour
D1_11	M1-Mechanical Prop [N/mm ²]	Vector (dim.3) reals of	Prevailing typology MECHANICAL PARAMETERS EVALUATION of the MASONRY (3 fields for compressive strength, shear strength and modulus of elasticity) [N/mm ²]
D3	MQI-Average	real (0,...,1)	Average MQI - Overall average quality index to be used as a parameter of the vulnerability model (weighted average between the 2 types of masonry of the minimum die and MQI indices). If not available, the reference average value is assigned

8.2.1.12. FOUNDATION

Attribute	Alias	Value Type	Definition
OBJECTID		ObjectID	ID of the element in the table
ConstructionRef		integer	Reference to the CONSTRUCTION table
E1	Site Morphology	integer (1,...,4)	Site morphology 1=crest 2=slope 3=mid slope 4=plan
E2	Foundations	integer (1,...,4)	Damage to the foundations 1=Absent 2=produced by earthquake 3=worsened 4=pre-existent

8.2.1.13. DAMAGE

Attribute	Alias	Value Type	Definition
OBJECTID		ObjectID	ID of the element in the table
ConstructionRef		integer	Reference to the CONSTRUCTION table
F1_2	Damage Floors	Integer (0,333)	Damage to the structural components: floors Score as F1-1
F1_3	Damage Stairs	Integer (0,333)	Damage to the structural components: stairs Score as F1-1
F1_4	Damage Roof	Integer (0,333)	Damage to the structural components: roof Score as F1-1
F1_5	Damage Partition	Integer (0,333)	Damage to the structural components: infill-partitions Score as F1-1
F1_6	Pre-existing Damage	Integer (0,333)	Damage to the structural components: Pre-existing damage Score as F1-1

Attribute	Alias	Value Type	Definition
F1_1	Damage Vertical Structures	Integer (0..333)	Damage to the structural components: vertical structures Code consisting of 3 integers: The first position (hundreds) describes the extension of very heavy damage, The second position (tens) describes the extension of medium severe damage, the third one (units) describes the extension of slight damage Extension is described by the following code 0= no damage or negligible damage 1=extension >2/3 2=1/3<extension <2/3 3=extension <1/3 (ordered as in AEDES form) e.g. 002 (or 2)=slight damage with extension between 1/3 and 2/3 110 = very heavy damage with extension<1/3 and medium damage with extension<1/3
F2_1	Damage Coverings	Integer (0,1)	Damage to the no structural components: falling of plaster, coverings, false ceilings 0=absent 1=present
F2_2	Damage Tiles Chimneys	Integer (0,1)	Damage to the no structural components: falling of tiles, chimneys Score as F2-1
F2_3	Damage Eaves	Integer (0,1)	Damage to the no structural components: falling of eaves, parapets Score as F2-1
F2_4	Damage Objects	Integer (0,1)	Damage to the no structural components: falling of internal or external objects Score as F2-1
F2_5	Damage Hydraulic	Integer (0,1)	Damage to the no structural components: damage to hydraulic or sewage systems Score as F2-1
F2_6	Damage Electric/gas	Integer (0,1)	Damage to the no structural components: damage to electrical or gas systems Score as F2-1
F3	Usability Classification	integer (1,...,6)	Usability classification 1=usable building 2= usable with countermeasures (temporary unusable) 3=partially unusable 4=requiring more detailed investigation (temporary unusable) 5=unusable 6=unusable for external risk

8.2.1.14. SOCIAL_CULTURAL_HISTORICAL_VALUE

Attribute	Alias	Value Type	Definition
OBJECTID		ObjectID	ID of the element in the table
ConstructionRef		integer	Reference to the CONSTRUCTION table

G1	Historical Value	Real (0..10)	Historical value
G2	Cultural, aesthetic and symbolic value	Real (0..10)	Cultural, aesthetic and symbolic value
G3	Social value	Real (0..10)	Social value
G4	Religious, Spiritual value	Real (0..10)	Religious, spiritual and commemorative value
G5	Recreational value	Real (0..10)	Recreational value
G6	Other value	Real (0..10)	Other value
G7	Socio-Cultural Index	Real (0..1)	Index value for estimating a loss. If this information is missing, the reference value 0.5 is assigned

8.2.1.15. ELEVATION

Attribute	Value Type	Definition
OBJECTID	ObjectID	ID of the element in the table
ElevationReferenceRef	ElevationReferenceValue	Refer to ELEVATION_REFERENCE_VALUE table
ElevationValue	Real	Value of the elevation.

8.2.1.16. ELEVATION_REFERENCE

Attribute	Value Type	Definition
OBJECTID	ObjectID	ID of the element in the table
ElevationLabel	ElevationReferenceValue_Label	see ElevationReferenceValue table
ElevationDescription	ElevationReferenceValue_Description	see ElevationReferenceValue table

8.2.1.17. HEIGHT_ABOVE_GROUND

Attribute	Value Type	Definition
OBJECTID	ObjectID	ID of the element in the table
BuildingRef	integer	Reference to the BUILDING table
HeightReference	ElevationReferenceValue	Element used as the high reference.
LowReference	ElevationReferenceValue	Element as the low reference.
HeightStatus	HeightStatusValue	The way the height has been captured.
Value	Length	Value of the height above ground.

8.2.1.18. HEIGHT_STATUS

Attribute	Value Type	Definition
OBJECTID	ObjectID	ID of the element in the table
HeightStatusLabel	HeightStatusValue_Label	see label in HeightStatusValue table
HeightStatusDescription	HeightStatusValue_Description	Refer to HeightStatusValue table

8.2.1.19. HORIZONTAL_GEOMETRY_REFERENCE

Attribute	Value Type	Definition
OBJECTID	ObjectID	ID of the element in the table
HorizontalGeometryLabel	HorGeoReferenceValue_Label	refer to HorGeoReferenceValue table
HorizontalGeometryDesc	HorGeoReferenceValue_Description	refer to HorGeoReferenceValue table

8.2.1.20. MEASURE

Attribute	Value Type	Definition
OBJECTID	ObjectID	ID of the element in the table
MeasureType	integer	Reference to the MEASURE_TYPE table
MeasurementDate	Date	Date of the measure
ConstructionRef	integer	Reference to the CONSTRUCTION table
Marker3D	integer	Reference to the Marker3Dvalue table

8.2.1.21. MEASURE_TYPE

Attribute	Value Type	Definition
OBJECTID	ObjectID	ID of the element in the table
MeasureTypeLabel	MeasureTypeValue_Label	see label in MeasureTypeValue table
MeasureTypeDescription	MeasureTypeValue_Description	Refer to MeasureTypeValue table

8.2.1.22. BuildingNatureValue

Label	Description
arch	A man-made structure in the form of an arch.
bunker	A facility, partly underground, intended for or used by the military either for location of command/control centres or for troop encampment.
canopy	An overhead roof providing shelter to things below. Canopies may be free standing frameworks over which a covering is attached or may be linked or suspended to the outside of a building.
caveBuilding	A space hosting human or economic activity which is usually enclosed within rock with the addition of man-made exterior walls and which may contain structures comparable to the interior structures of freestanding buildings.
chapel	A Christian place of worship, usually smaller than a church.
castle	A large ornate or fortified building usually constructed for the purpose of a private residence or security.
church	A building or structure whose primary purpose is to facilitate the catholic or orthodox cult.
dam	A permanent barrier across a watercourse used to impound water or to control its flow.

Label	Description
greenhouse	A building that is often constructed primarily of transparent material (for example: glass), in which temperature and humidity can be controlled for the cultivation and/or protection of plants.
lighthouse	A tower designed to emit light from a system of lamps and lenses.
mosque	A building or structure whose primary purpose is to facilitate the Muslim cult.
shed	A building of light construction, which usually has one or more open sides, that is typically used for storage.
silo	A large storage structure, generally cylindrical, used for storing loose materials.
stadium	A place or venue for sports, concerts or other events and consists of a field or stage either partly or completely surrounded by a structure designed to allow spectators to stand or sit and view the event.
storageTank	A container usually for holding liquids and compressed gases.
synagogue	A building or structure whose primary purpose is to facilitate the is religious cult.
temple	A building or structure whose primary purpose is to facilitate the meeting of a religious sect.
tower	A relatively tall, narrow structure that may either stand alone or may form part of another structure.
windmill	A building which converts the energy of the wind into rotational motion by means of adjustable sails or blades.
windTurbine	A tower and associated equipment that generates electrical power from wind.
Column	An upright pillar, typically cylindrical, supporting an arch, entablature, or other structure or standing alone as a monument.
Monastery	A building occupied by a community of monks living under religious vows.
Obelisk	A tapering stone pillar, typically having a square or rectangular cross section, set up as a monument or landmark.
Palace	A large and impressive building forming the official residence of a ruler, pope, archbishop, etc.
Trilithon	A megalithic structure consisting of two upright stones and a third across the top as a lintel.
Triumphal Arch	A monumental structure in the shape of an archway with one or more arched passageways
Groups buildings	Groups of separate or connected buildings
Historical nucleus	Historical nucleus
Monumental sculpture and painting	Works of monumental sculpture and painting
Gardens	Parks and gardens
nature human work	Combined work of nature and humankind
Archaeological finds	Archaeological finds
Archaeological sites	Archaeological sites

8.2.1.23. ConditionOfConstructionValue

Label	Description
declined	A construction cannot be used under normal conditions, though its main elements (walls, roof) are still present.
demolished	A construction has been demolished. There are no more visible remains.
functional	A construction is functional.
projected	A construction is being designed. Construction has not yet started.
ruin	The construction has been partly demolished and some main elements (roof, walls) have been destroyed. There are some visible remains of construction.
underConstruction	A construction is under construction and not yet functional. This applies only to the initial construction of the construction and not to maintenance work.
underRenovation	A construction is under renovation and not yet functional.

8.2.1.24. CurrentUseValue

Label	Description
residential	A building (or building component) is used for residential purpose.
individualResidence	A building (or building component) hosts only one dwelling.
collectiveResidence	A building (or building component) hosts more than one dwelling.
twoDwellings	A building (or building component) hosts two dwellings.
moreThanTwoDwelling	A building (or building component) hosts at least 3 dwellings.
residenceForCommunities	A building (or its component) hosting a residence for communities.
agriculture	A building (or building component) is used for agricultural activities.
industrial	A building (or building component) is used for secondary sector activities (industrial).
commerceAndServices	A building (or building component) is used for any service activities. This value addresses the buildings and building components dedicated to tertiary sector activities (commercial and services).
office	A building (or building component) hosts offices.
trade	The building (or building component) hosts trade activities.
publicServices	A building (or building component) hosts public services. Public arteria services provided for the benefit of the citizens.
ancillary	A building (or building component) of small size that is used only in connection with another larger building (or building component) and generally does not inherit the same function and characteristics as the building (or building component) it is linked to.
religious	A building (or building component) hosts religious activities.

8.2.1.25. ElevationReferenceValue

Label	Description
aboveGroundEnvelope	The elevation has been captured at the level of the maximum extent of the above ground envelope of the construction.
bottomOfConstruction	The elevation has been captured at the bottom of the usable part of the construction.
entrancePoint	The elevation has been captured at the entrance of the construction, generally the bottom of entrance door.
generalEave	The elevation has been captured on one of the meeting lines between the roof and the walls.
generalGround	The elevation has been captured on one of the meeting lines between the construction and the ground.
generalRoof	The elevation has been captured anywhere on the roof.
generalRoofEdge	The elevation has been captured on one of the roof edges.
highestEave	The elevation has been captured on the highest meeting line between the roof and the walls.
highestGroundPoint	The elevation has been captured on the highest point of the meeting lines between the construction and the ground.
highestPoint	Elevation has been captured at the highest point of the construction, including the installations, such as chimneys and antennas.
highestRoofEdge	The elevation has been captured at the highest roof edge level of the construction.
lowestEave	The elevation has been captured on the lowest meeting line between the roof and the walls.
lowestFloorAboveGround	The elevation has been captured at the level of the lowest floor above ground.
lowestGroundPoint	The elevation has been captured on the lowest point of the meeting lines between the construction and the ground.
lowestRoofEdge	The elevation has been captured at the lowest roof edge level of the construction.
topOfConstruction	The elevation has been captured at the top level of the construction.

8.2.1.26. HeightStatusValue

Label	Description
estimated	The height has been estimated and not measured.
measured	The height has been (directly or indirectly) measured.

8.2.1.27. HorGeoReferenceValue

Label	Description
aboveGroundEnvelope	The building horizontal geometry has been captured using the above ground envelope of the building, i.e. the maximum extent of the building above ground.
combined	The building horizontal geometry has been obtained from the combination of the geometries of its building parts with the geometries of the building parts using different horizontal geometry references.
entrancePoint	The building geometry is represented by a point located at the entrance of the building.
envelope	The building horizontal geometry has been captured using the whole envelope of the building, i.e. the maximum extent of the building above and underground.
footPrint	The building horizontal geometry has been captured using the footprint of the building, i.e. its extent at ground level.
lowestFloorAboveGround	The building horizontal geometry has been captured using the lowest floor above ground of the building.
pointInsideBuilding	The building horizontal geometry is represented by a point located within the building.
pointInsideCadastralParcel	The building horizontal geometry is represented by a point located within the parcel the building belongs to.
roofEdge	The building horizontal geometry has been captured using the roof edges of the building.

8.2.1.28. MeasureTypeValue

Label	Description
OTH	Unknown
3DM	3D Model
PDF	PDF file
TXT	Text file
PIC	Picture
LSC	Laser Scanning
IRT	IR Thermographic image
CAD	Technical drawing CAD
STM	Static monitoring
DYM	Dynamic monitoring
HYM	Hyperspectral analysis on construction materials
SCL	Sclerometer Test
SNC	Sonic Test
UTS	Ultrasonic test

8.2.2. Tables for the *OBJECT* database

8.2.2.1. OBJECT

Attribute	Value Type	Definition
OBJECTID	Object ID	
SheetRef	Integer	Reference to the existent identification
StorageAreaRef	Integer	Reference to STORAGE_AREA table defining a container
ShelterAreaRef	Integer	Reference to STORAGE_AREA table defining a shelter in emergency phase
CurrentStorageArea	text	Current Storage Sector
HistoricStorageArea	text	Historic Storage Sector
Title	text	title or subject of the artefact
Quantity	Integer	Quantity of the elements
Description	Text	Description of the object\artefact
Author	Text	Author of the object\artefact
Diameter	Double	diameter dimension in cm
Width	Double	width dimension in cm
Depth	Double	depth dimension in cm
Height	Double	height dimension in cm
Category	Integer (1..3)	1 movable 2 immovable (3 intangible)
DisciplinarySector	Integer (1..10)	1 archaeological 2 architecture 3 landscape 4 ethnoanthropological 5 photographic 6 musical 7 naturalistic 8 numismatists 9 scientific-technological 10 historical-artistic
Artefact	TypeArtefact	1 painting 2 wall painting 3 sculpture 4 religious object 5 decoration
Officer	text	Officer
Note	text	Note
Date	date	Date from which the object is in the current position

8.2.2.2. STORAGE_AREA

Attribute	Value Type	Definition
OBJECTID	Object ID	
CityRef	Integer	Reference to the city
ConstructionRef	integer	Reference to the CONSTRUCTION table
Name	text	Name of the storage area
Longitude	double	Longitude
Latitude	double	Latitude
Shelter	Integer (1..2)	1 Conventional storage 2 Shelter (storage in emergency phase)

8.2.2.3. EVENT

Attribute	Value Type	Definition
OBJECTID	Object ID	
ArtefactRef	Integer	Reference to the OBJECT table
Purpose	text	Purpose of the movement of the objects
Description	text	Description of the event for which the object will be moved
DateStart	date	Start date of the event
DateEnd	date	End date of the event
ExternalResponsible	text	Responsible internal to the city\stakeholder
InternalResponsible	text	Responsible internal to the city\stakeholder

8.2.2.4. INTERVENTION

Attribute	Value Type	Definition
OBJECTID	Object ID	
ArtefactRef	Integer	Reference to the OBJECT table
Rating	integer (1..5)	Rating assigned to the intervention on the artefact: 1 very low 2 low 3 medium 4 high 5 very high
Urgency	integer (1..5)	Urgency assigned to the intervention on the artefact: 1 very low 2 low 3 medium 4 high 5 very high
EvaluationDate	Date	Evaluation Date of the intervention

Attribute	Value Type	Definition
DateStart	Date	Start date of the intervention
DateEnd	Date	End date of the intervention
Responsible	Text	Responsible of the intervention
Description	Text	Description of the intervention

8.2.2.5. ATTACHMENT

Attribute	Value Type	Definition
OBJECTID	Object ID	
TableName	TableName	Reference to the table: - OBJECTS - INTERVENTION -...
ID_ref	Integer	Refence to the ID of the table TableName
File	Text	Path + filename